

Vortical Enclosure Dynamics, Vol. 4

The Differential Horizon: Divergence, Renormalization, and the Boundary of Generation

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Abstract

We propose the Differential Horizon as a structural framework for reinterpreting divergence and renormalization in modern physics. Rather than treating divergence as a technical failure to be corrected, we argue that it constitutes a signal of boundary contact—an indication that a physical theory has approached the limit at which the generated structure of spacetime gives way to the un-generated domain, where no descriptive framework applies.

From this perspective, renormalization is not an operation that eliminates infinity, but one that converts boundary contact into a finite internal representation. The speed-of-light limit, the Planck-scale breakdown of spacetime continuity, and black hole singularities are re-positioned as different modes of approach to the same boundary, rather than as independent problems requiring separate solutions.

This framework does not propose a unified theory of interactions. Instead, it identifies a prior structural layer: the condition under which any physical theory can hold. The Differential Horizon Principle states that the un-generated domain is in principle unreachable, that approach to its boundary produces divergence, and that renormalization is the mechanism by which that contact is absorbed into theory.

VED (Vortical Enclosure Dynamics) is positioned within this framework as a theory of the generation boundary—not a description of existence, but of the conditions under which existence begins.

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1 Introduction

The problem of renormalization has stood at the center of modern physics for decades. Quantum field theory produces divergent quantities mid-calculation; renormalization absorbs these into observable definitions and yields predictions of extraordinary precision. Yet the physical meaning of this procedure has never been fully explained.

This paper takes that explanatory gap as its starting point. We do not treat divergence as a mathematical artifact to be managed, but as a structural signal: an indication that physical theory is approaching a boundary that it cannot cross.

We refer to this boundary as the Differential Horizon. It is the boundary at which difference first arises—the transition between the *un-generated* domain, where no physical description applies, and the *generated* domain, where spacetime, fields, and laws come to hold.

Three claims organize this paper. First, divergence is not failure but *boundary contact*: it appears when theory reaches the outer edge of its descriptive domain. Second, renormalization is not elimination but conversion: it translates the influence of that contact into a finite internal representation. Third, the speed-of-light limit, the Planck-scale breakdown, and black hole singularities are not independent problems but different modes of approach to the same boundary.

This framework does not propose a new unified theory. It identifies a prior structural layer: the condition under which any physical theory can hold, and makes that condition explicit through the *Differential Horizon Principle*.

VED (Vortical Enclosure Dynamics) is positioned within this framework as a theory of the generation boundary itself. It does not describe existence; it describes the conditions under which existence begins.

The paper proceeds as follows. Section 2 examines the unease of renormalization and reframes it as a structural clue. Sections 3–4 establish that physics consistently describes only post-generated structure and that the un-generated domain is in principle indescribable. Section 5 defines the Differential Horizon. Sections 6–8 develop the reinterpretation of divergence and renormalization. Sections 9–10 replace major theories and unsolved problems within the framework. Section 11 states the Differential Horizon Principle. Section 12 positions VED. Section 13 concludes.

2 The Unease of Renormalization

Renormalization is one of the most successful procedures in modern physics. Quantum field theory produces divergent quantities during calculation; renormalization absorbs these divergences into observable definitions and yields predictions that

agree with experiment to extraordinary precision.

Yet this success conceals a fundamental unease.

The procedure works with remarkable accuracy, yet its conceptual meaning remains unresolved. A theory that produces infinities mid-calculation is nevertheless able to describe physical reality correctly after appropriate redefinition.

This produces a peculiar situation.

A theory breaks down once, yet after modification, it succeeds.

Why does a theory that diverges still succeed?

This question is rarely pursued at a foundational level. Renormalization has been formalized and refined, and its mathematical structure is well understood. However, its physical meaning remains unclear.

What, precisely, is renormalization doing?

Why do divergences appear? And why does removing them yield correct predictions?

No unified interpretation of these questions has been established.

In this paper, we take this unease seriously.

Rather than treating renormalization as a problem to be resolved, we treat it as a structural clue.

The appearance of divergence may not indicate a failure of theory, but a contact with something beyond its domain.

From this perspective, renormalization is not merely a technical operation. It may be a procedure that converts the influence of that contact into a form that can be handled within the theory.

This reinterpretation will be developed throughout this volume.

Divergence is not removed; it is transformed.

And what appears as infinity may be the signal of a boundary.

3 Physics Describes Only the Past

The preceding discussion suggested that divergence may not be a mere failure of theory, but a signal indicating that theory is approaching a boundary.

To understand this possibility, we must first examine the structure on which physical theory is built.

Modern physics consists of multiple frameworks: classical mechanics, relativity, quantum mechanics, and quantum field theory. Each applies to different scales and conditions, yet all share a common structure.

They assume that time and space are given.

Physics takes a state defined at a moment in time and describes how it evolves. Or it defines a field in space and describes its behavior through equations.

In all cases, the object of description is something that already exists.

Particles, fields, energy, spacetime— these are given in advance, and theory describes their relations and changes.

This structure is so natural that it often goes unnoticed.

Yet it has a crucial implication.

Physics describes only the past.

The past here does not mean what has elapsed in time. It means what has been fixed into a form that can be described.

What can be observed, recorded, and expressed in equations is always something that has already been stabilized as structure.

Physical theory takes these stabilized structures as input and describes the relations between them.

Even when predicting the future, the prediction proceeds from already fixed initial conditions.

In this sense, physics always operates within the domain of the past.

This structure is not accidental.

Physical theory presupposes several basic elements: time, space, states, and observable quantities.

These are not derived within theory; they are presupposed.

All equations, symmetries, and conservation laws are defined on the basis of these given structures.

From this perspective, physics consistently describes the domain after generation has occurred.

What, then, lies outside this domain?

A domain where time has not yet acquired meaning. Where space does not yet hold. Where states are not defined, and no observable quantities exist.

How does theory respond to such a domain?

The answer may be divergence.

Divergence appears precisely when theory is pushed beyond the domain in which its presupposed structures remain valid.

Equations no longer yield finite values. Quantities become undefined. The framework itself ceases to function.

This is not merely computational failure.

It is the response of theory when it approaches the limit of its own conditions of applicability.

Physics does not describe existence.

It describes the structure after generation.

From this view, divergence is not a defect of theory, but an indicator of its boundary.

Beyond that boundary lies a domain that cannot be described within the existing framework.

Not merely unknown, but in principle indescribable.

The next section examines the nature of this boundary more closely.

Supplement: What Is Pastification? — Physics as Texture

Let us restate the concept of *pastification* more concretely.

The past is not merely what has elapsed in time. It is what can be handled as a tangible form.

Color, temperature, weight, sensation— all arise from micro-level details accumulating and being compressed at some scale.

A continuous physical quantity called wavelength rises up as the color red or blue. The microscopic structure of woven thread appears as the tactile quality of smooth or soft.

In neither case do the raw details appear directly. They are compressed and crystallized as texture, and only then perceived as quality.

Symmetry and conservation laws in physics share this structure. The accumulation of individual relations appears, viewed from a certain distance, as an invariant structure. It is an integral of relations—a texture born through compression.

In this sense, the past is the accumulated crystallization of relations as texture.

What, then, is the *pastification* of the future?

The future in principle contains possibilities that can diverge without limit. These cannot be handled as they are.

When physics predicts the future, it extracts from this diverging possibility space only the quality that can be handled as texture through computation, and places it on the same footing as the past.

Pastification of the future is, in other words, the operation by which endlessly expanding possibilities are crystallized into texture.

And the mathematical procedure of that operation is renormalization.

Renormalization does not eliminate divergence. It extracts micro-scale information as texture at a coarser scale— compressing it into a form that can be handled as texture.

That is why renormalized theory describes reality well: because it has been compressed consistently as texture.

From this perspective, the meaning of the *Differential Horizon* becomes sharper.

As one approaches the horizon, the very foundation for compression is lost. Structure that could be extracted as texture disappears. Texture can no longer be constituted.

That is what appears as divergence.

The *un-generated* domain is the domain prior to the emergence of texture. Not even the individual points of pointillism yet exist— neither color nor tactile quality has yet emerged.

Physics holds only within the world of texture.

Of what lies before it, nothing can be said.

4 Why the Un-generated Cannot Be Handled

The previous section established that physical theory consistently describes the domain after generation has occurred.

This naturally raises a question.

Why can the domain prior to generation not be described?

One might initially assume that this domain is simply unknown, and that a more advanced theory will eventually describe it.

However, the issue here is not merely one of incomplete knowledge.

It concerns the form of description itself.

Physical theory presupposes several basic structures: time, space, states, and observable quantities. These are taken as given, and laws are defined on the basis of them.

The *un-generated* domain fits none of these presuppositions.

If time has not yet come to hold, the question of “when” has no meaning. If space has not yet come to hold, the question of “where” does not arise. If states are not defined, quantities cannot be assigned. If observable quantities do not exist, theory has no object.

From this, a crucial point follows.

The *un-generated* is not merely unknown.

It is not something that can, in principle, be described.

It does not fit the form of description itself.

No matter how precise a theory becomes, as long as it presupposes structures such as time and space, it cannot reach this domain.

This is not a limitation of current theory, but a constraint imposed by the conditions under which theory holds.

Because theory presupposes these structures,
it cannot extend beyond them.

At this point, the significance of divergence becomes clearer.

Theory cannot describe the *un-generated* domain directly. But it can approach its boundary.

As it does so, quantities cease to be defined, and divergence appears.

This is not accidental.

It indicates that a discontinuity exists between the domain that can be described and the domain that cannot.

However, this boundary is not a sharp line.

It is a transition band of finite thickness.

Physical theories always operate within finite precision and scale. The *generated* and the *un-generated* cannot be cleanly separated.

Between them exists a region in which the descriptive framework gradually loses validity.

Within this region, theory does not immediately fail, but becomes unstable, and eventually produces divergence.

From this perspective, the following statement becomes natural.

Divergence is not reaching the *un-generated* domain.

It is contact with what lies just before it.

This contact does not allow direct description. But it cannot be denied.

Theory approaches, responds, and then breaks down. The manner of this breakdown itself is the only available indication of the boundary.

Thus, the *un-generated* domain does not appear within theory directly. It appears only through its boundary.

The next section defines this boundary explicitly.

It is not introduced as an abstract limit, but as a structure with physical meaning: the *Differential Horizon*.

5 The Differential Horizon

From the preceding discussion, one necessity has become clear.

Physical theory consistently describes structure only after generation. The *un-generated* domain itself cannot, in principle, be described.

Yet theory approaches its boundary, and responds with divergence.

At this point, the introduction of a new concept becomes necessary.

The *Differential Horizon* is the boundary at which difference first arises.

In the *un-generated* domain, there is no difference. No distinction, no change, no position, and no time can be defined.

All points are indistinguishable; no structure can be extracted.

When generation occurs, the situation changes completely.

Difference arises, distinctions become possible, space emerges, and time begins to flow.

Physical laws hold only on the basis of this difference.

The *Differential Horizon* is therefore not merely a boundary.

It is the condition under which physics becomes possible.

However, this boundary is not a sharp dividing surface.

It is a transition band of finite thickness.

This transition band fluctuates. It is not uniform, and is influenced by energy distribution and gravitational effects.

In this sense, the *Differential Horizon* is not a fixed geometric surface, but a dynamically deforming region.

It is crucial to note the following.

The *Differential Horizon* cannot be reached as an object of theory.

No theory can describe the *un-generated* domain itself.

However, theory can approach its boundary.

And at the limit of that approach, divergence appears.

From this perspective, divergence takes on a precise meaning.

Divergence is the signal of contact with the *Differential Horizon*.

Equations do not fail arbitrarily. They indicate that they are reaching the outer edge of their domain.

From this view, phenomena previously treated as separate can be understood within a single structure.

Divergence in quantum field theory, breakdown at the Planck scale, the speed-of-light limit, and black hole singularities—

all are different expressions of approach toward the *Differential Horizon*.

Not different theories.

Different paths of approach.

All physical theories, though differing in method, approach this same boundary. The *Differential Horizon* lies outside all theories. Yet all theories move toward it.

Physics operates inside the horizon.

What lies outside cannot be described.

What can be described is only the response when approaching it.

The *Differential Horizon* is not a concept for describing existence.

It is a concept for describing the onset of generation.

Figure 1 illustrates the overall structure: the *un-generated* domain, the *Differential Horizon* as transition band, and the *generated* domain in which texture, symmetry, and conserved quantities emerge through compression and coarse-graining.

The next section examines how this approach occurs in concrete terms, through two distinct modes: spatial convergence and temporal convergence.

6 Two Paths of Approach

The *Differential Horizon* cannot be reached directly. But it can be approached.

This approach takes two distinct forms.

Spatial convergence and temporal convergence are dual modes of approach to the same boundary.

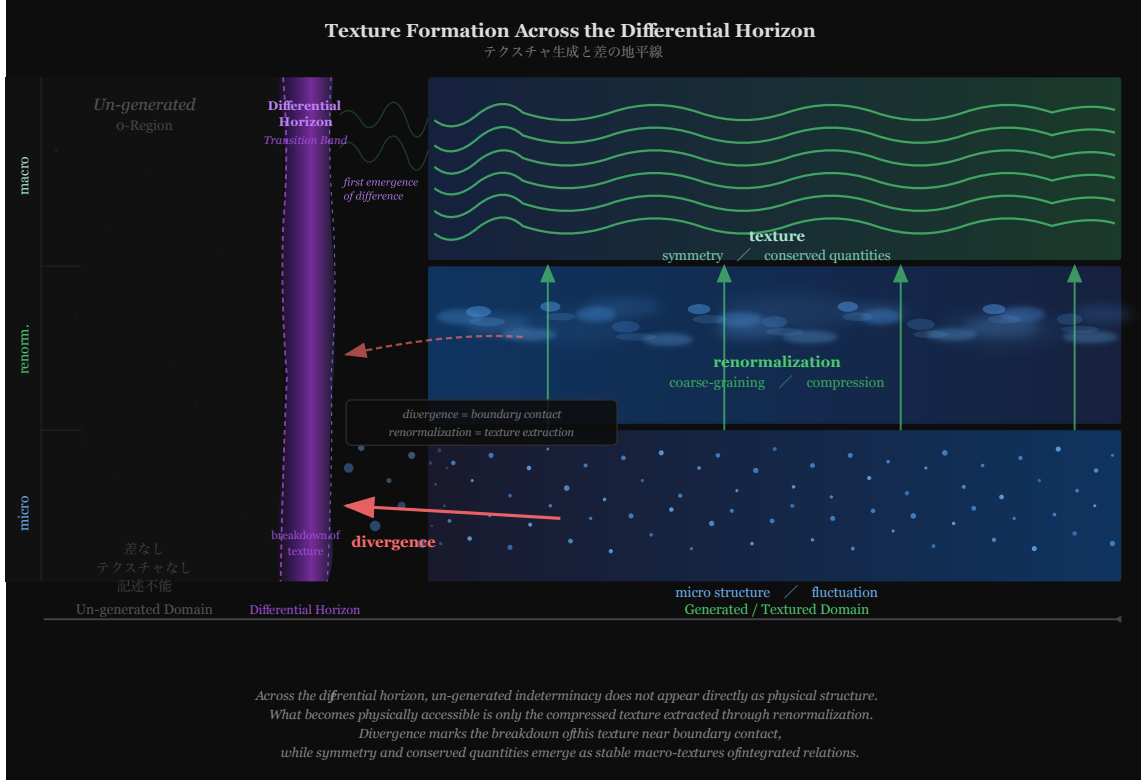


Figure 1: Texture formation across the *Differential Horizon*. The *un-generated* domain (left) contains no difference, no structure, and no describable form. The transition band (center) is the *Differential Horizon*. The *generated* domain (right) is organized into three layers: micro-structure and fluctuation (bottom), renormalization and coarse-graining (middle), and stable texture, symmetry, and conserved quantities (top). Divergence marks the breakdown of texture near *boundary contact*. *The world appears not as raw microstructure, but as compressed texture.*

6.1 Spatial Convergence

Approaching ever smaller scales leads into the quantum domain.

Particles lose classical localization, fields fluctuate strongly, and quantities such as position and momentum cease to be simultaneously definable.

As the scale decreases further, theory gradually loses stability.

Quantum fluctuations intensify, and calculations develop divergences.

At the Planck scale, the continuity of spacetime itself is expected to break down.

Distance and position lose meaning.

Reducing scale is approaching the *Differential Horizon*.

This path is described by quantum theory and quantum field theory. Divergence appears frequently, and renormalization becomes indispensable.

6.2 Temporal Convergence

The second path arises through increasing velocity.

In relativity, as velocity approaches the speed of light, time dilation becomes extreme.

At the limit, time ceases to be definable.

Simultaneously, length contraction eliminates spatial extension along the direction of motion.

Approaching the speed of light is approaching the *Differential Horizon*.

This path is described by relativity. Energy diverges toward infinity, and a body with finite mass cannot reach the speed of light. The structure of theory itself does not permit it.

6.3 Unified Structure

Though these two paths appear fundamentally different, they share a common structural feature.

Both paths lead to the loss of definability of spacetime structure.

In spatial convergence, fluctuation destroys localization.

In temporal convergence, extreme motion destroys temporal duration.

In both cases, difference itself ceases to be well-defined.

Different paths.

Same loss of definability.

This is not coincidence.

The problem of quantum gravity, viewed from this perspective, is not simply a matter of integrating two theories. It is a matter of understanding the structure by which two paths of approach converge on the same boundary.

They are not different domains, but different trajectories within the same structure.

All approaches toward the *Differential Horizon* lead to the same condition: the breakdown of the structures that make physical description possible.

This structure is illustrated in Figure 2.

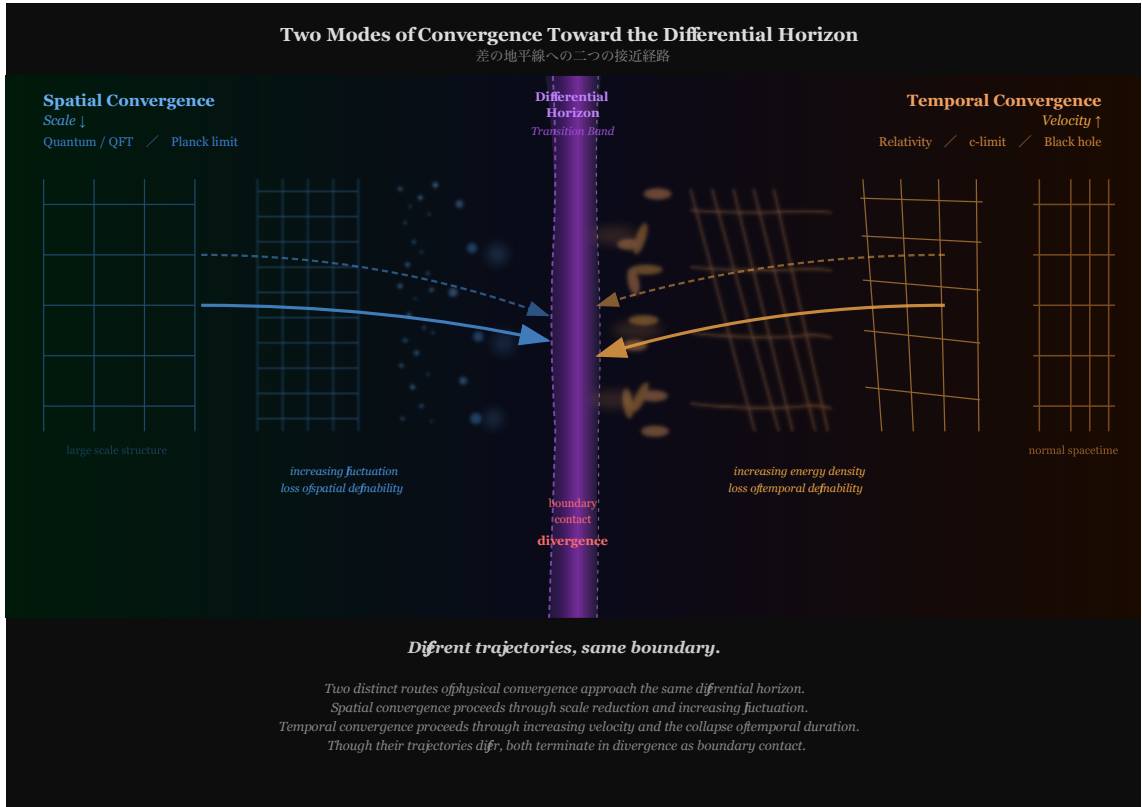


Figure 2: Two modes of convergence toward the *Differential Horizon*. Spatial convergence (left, blue) proceeds through scale reduction and increasing fluctuation toward the Planck limit. Temporal convergence (right, orange) proceeds through velocity increase and the collapse of temporal duration toward the speed-of-light limit. Both routes terminate in divergence as *boundary contact*. *Different trajectories, same boundary.*

The next section redefines divergence formally in light of this convergent structure.

7 Redefining Divergence

From the preceding discussion, one conclusion becomes unavoidable.

A physical theory cannot describe the *un-generated* domain itself. But it can approach its boundary.

At the limit of that approach, divergence necessarily appears.

Divergence is not an error to be removed.

Traditionally, divergence has been treated as a failure of theory.

However, this interpretation is not required.

Divergence is the moment at which a theory reaches the limit of its domain and registers contact with the *Differential Horizon*.

This is not metaphorical.

Divergence appears whenever a theory is pushed toward its extreme limits: when spatial scale approaches zero, when energy approaches infinity, or when gravitational curvature becomes extreme.

Regardless of the specific physical context, the same structural pattern is observed.

Divergence necessarily appears at the boundary of the describable domain.

This is not a coincidence.

It indicates that the theory is encountering a domain that lies beyond its descriptive capacity.

What Divergence Does Not Reveal

Importantly, divergence does not provide information about what lies beyond that boundary.

Divergence does not reveal the structure beyond the boundary.

It only reflects the limit of the theory itself.

Thus, divergence is not a means of crossing the boundary.

Divergence does not provide a means of access beyond the boundary.
It marks the impossibility of extending the theory further.

From this perspective, divergence is not failure.

Divergence is a contact signal.

It indicates that the theory has reached the edge of its domain and is responding to that contact.

Unification of Previously Separate Phenomena

This reinterpretation unifies phenomena that have traditionally been treated separately.

- Divergence in quantum field theory
- Singularities in general relativity
- The breakdown of spacetime at the Planck scale
- Extreme behavior near the speed-of-light limit

These are not independent anomalies.

They are different appearances of contact with the *Differential Horizon*.

They differ in path and in form, but share the same underlying structure: theory has reached the outer edge of its domain.

Implication for Renormalization

At this point, the meaning of renormalization begins to shift.

The response to this contact is not elimination, but transformation.

Rather than removing divergence, renormalization converts the effects of *boundary contact* into a form that can be handled within the theory.

Divergence and renormalization are two sides of the same phenomenon.

This will be examined in detail in the next section.

8 Reinterpreting Renormalization

The previous section redefined divergence as contact with the *Differential Horizon*.

From this perspective, the meaning of renormalization also changes fundamentally.

Renormalization is not an operation that removes divergence.

Instead,

Renormalization is the operation that converts *boundary contact* into finite internal representation.

Two Layers of Physical Quantities

In quantum field theory, quantities appearing in calculations often diverge.

However, observable quantities remain finite.

Renormalization connects these two layers.

Quantities containing divergences are absorbed into the definition of observables, and predictions emerge as finite values.

Conventionally, this is regarded as a technical procedure.

But from the perspective of the *Differential Horizon*, its structure becomes clear.

Theory cannot directly handle the information of *boundary contact*.

Divergence is the direct expression of that contact. But in that form, it cannot be consistently treated within the theory.

Thus, it is transformed.

The divergence itself does not disappear.

Only its influence is retained in a transformed form.

Renormalization as Mapping

Renormalization is not elimination.

It is mapping.

More precisely, it is a projection of *boundary contact* into a form that can be handled within the theory.

It is not removal, but projection into a lower-resolution representation.

This explains why renormalization succeeds.

Renormalization succeeds because it preserves the effects of *boundary contact* while maintaining internal consistency of the theory.

Scale Dependence and Effective Theory

Several known features of renormalization follow naturally from this interpretation.

Scale dependence appears because *boundary contact* depends on the scale at which the theory is defined.

Different scales correspond to different modes of *boundary contact*.

Effective field theory works because the influence of high-energy structure is encoded in low-energy observables.

A physical quantity is a quantity that encodes the history of *boundary contact*.

Structural Necessity

Thus, renormalization is not merely a correction procedure.

It is the fundamental mechanism by which a theory remains valid within its own limits.

Divergence indicates contact;
renormalization makes that contact visible.

Divergence and renormalization are not opposing operations,
but two aspects of a single structural process.

The next section, based on this perspective, shows how major physical theories are placed in relation to the *Differential Horizon*— not by their objects, but by their mode of approach.

9 Theoretical Re-placement

From the preceding discussion, the structure of the *Differential Horizon* and the meaning of divergence and renormalization have become clear.

From this perspective, modern physical theories can be reorganized.

Theories are not classified by their objects,
but by their mode of approach to the *Differential Horizon*.

All theories are positioned within a continuous structure of approach.

9.1 Quantum Field Theory

Quantum field theory approaches the *Differential Horizon* through spatial convergence.

At high energies, field fluctuations increase, and calculations exhibit divergence.

These divergences are handled through renormalization, allowing finite predictions.

Quantum field theory is positioned just before the horizon, within a domain strongly influenced by *boundary contact*.

9.2 Relativity

Relativity approaches the *Differential Horizon* through temporal convergence.

As velocity approaches the speed of light, time dilates and tends toward cessation.

Under strong gravitational fields, spacetime structure breaks down, and singularities appear.

Relativity approaches the same boundary through a different mode of convergence.

9.3 Black Holes

Black holes represent the most explicit manifestation of *boundary contact*.

At the event horizon, the structure of spacetime becomes inaccessible to external observers.

At the singularity, curvature diverges, and spacetime loses definability.

Black holes occupy the region where *boundary contact* becomes manifest.

9.4 String Theory

String theory emerges as an attempt to unify quantum field theory and gravity.

Its mathematical structure seeks to suppress divergence and maintain internal consistency.

String theory does not directly touch the boundary,
but explores coherence near the limit of approach.

9.5 Quantum Gravity

Quantum gravity is often described as the problem of unifying quantum theory and relativity.

From the perspective of the *Differential Horizon*, its meaning shifts.

Quantum gravity is not a missing theory,
but a structural problem of convergence.

It concerns how different paths of approach lead to the same boundary.

9.6 Unified Structure

These theories are not independent.

These theories do not contradict one another.
They occupy different positions within the same structure.

Quantum field theory approaches through scale. Relativity approaches through velocity and curvature.

Black holes mark explicit contact. String theory explores coherence near the limit.

Quantum gravity expresses the difficulty of understanding their convergence.

All theories are different responses to the same boundary.

Figure 3 maps these positions.

The next section shows how this unified view gathers the major unsolved problems of modern physics under a single structural principle.

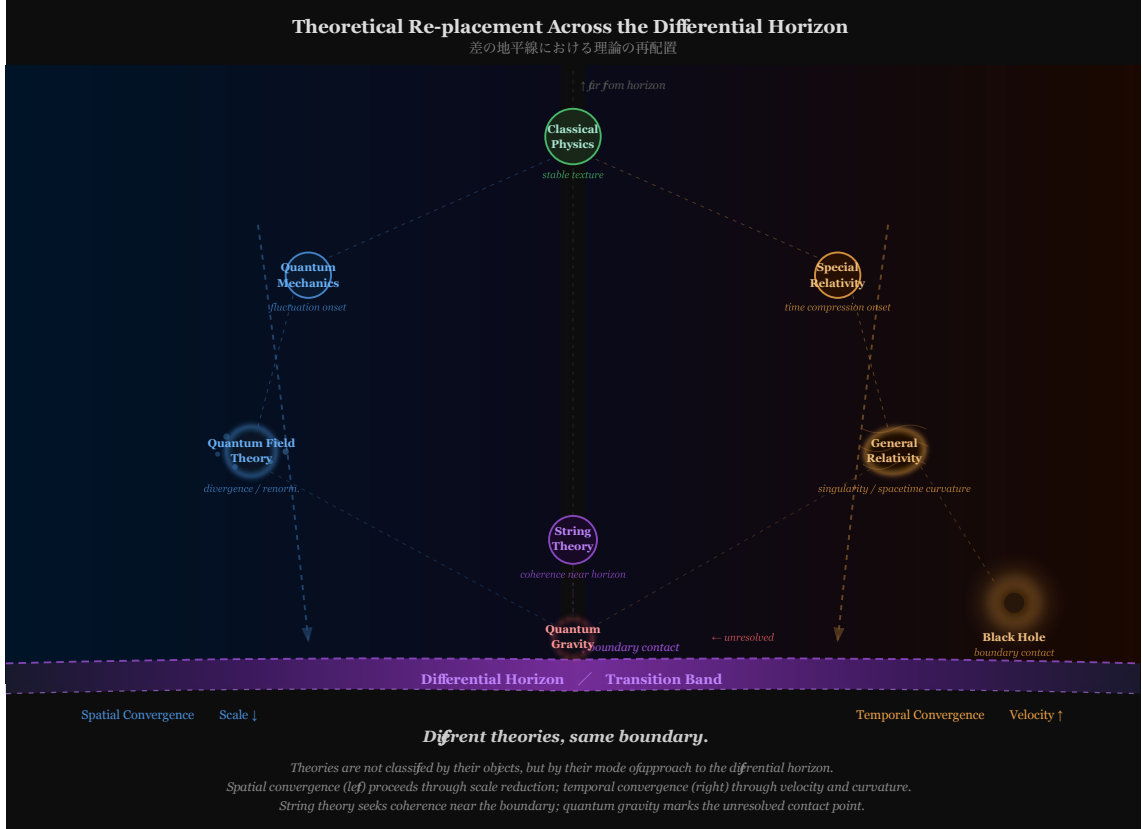


Figure 3: Theoretical re-placement across the *Differential Horizon*. Theories are organized not by their objects but by their mode of approach to the boundary. Spatial convergence (left) encompasses quantum mechanics and quantum field theory. Temporal convergence (right) encompasses special and general relativity. Black holes occupy the *boundary contact* region. String theory seeks coherence near the horizon. Quantum gravity marks the unresolved contact point. *Different theories, same boundary.*

10 Unification of Unsolved Problems

Modern physics contains many unsolved problems.

These are typically treated as independent issues belonging to different domains.

However, from the perspective of the *Differential Horizon*, a different view emerges.

The unsolved problems of modern physics are not independent problems, but different manifestations of a single structural condition.

Each appears when a theory is pushed to its limit of applicability.

10.1 Divergence in Quantum Field Theory

At high energies, quantum field theory exhibits divergences.

This reflects approach toward the *Differential Horizon* through spatial convergence.

10.2 Renormalization

Renormalization converts divergence into finite observable quantities.

It is the mechanism by which *boundary contact* is incorporated into theory.

10.3 Planck-Scale Breakdown

At the Planck scale, spacetime continuity loses meaning.

Distance and time cease to be definable.

This represents the limit of spatial convergence.

10.4 The Speed-of-Light Limit

As velocity approaches the speed of light, time dilation becomes extreme and temporal duration tends toward disappearance.

This represents the limit of temporal convergence.

10.5 Black Hole Singularities

Under extreme gravitational conditions, spacetime structure breaks down, and curvature diverges.

Black holes represent explicit *boundary contact*.

10.6 The Quantum Gravity Problem

Quantum gravity is often seen as the problem of unifying quantum theory and relativity.

From this perspective, its meaning shifts.

The difficulty of quantum gravity reflects the impossibility of extending post-generated frameworks beyond their domain.

10.7 Unified Interpretation

All these problems share a common structure.

All of them indicate approach toward the *Differential Horizon*.

They are not to be solved separately,
but to be re-placed within a unified structure.

These are not separate problems.

They are different modes of approach to the same boundary.

The next section states this structure as an explicit principle.

11 The Differential Horizon Principle

From the preceding discussion, a fundamental structure emerges.

Differential Horizon Principle:

The *un-generated* domain is in principle unreachable; approach to its boundary produces divergence; and renormalization converts that contact into finite representation.

This principle can be expressed in three propositions.

1. The *un-generated* domain is unreachable.
2. Divergence appears at the boundary of approach.
3. Renormalization converts *boundary contact* into finite form.

The Chain of Necessity

These propositions are not independent.

They form a necessary structural chain.

Because the *un-generated* domain cannot be described, approach to its boundary appears as divergence.

Because divergence cannot be handled directly, it is transformed through renormalization.

Physical theory operates only within this structure.

What the Principle Reveals

This principle does not introduce a new theory.

It reveals the structural constraint already present in existing theories.

Divergence, renormalization, singularities, and scale limits are not anomalies. They are expressions of this principle.

The *un-generated* is not an object.

It appears only as a boundary.

Scope and Limit

The *Differential Horizon Principle* does not explain existence.

It defines the condition under which explanation becomes possible.

It does not introduce new entities. It does not propose a unified theory of interactions. It does not describe what lies beyond the horizon.

What it does is identify the prior structural layer within which all physical description holds.

The *Differential Horizon Principle* is not a theory of what exists, but a principle of how existence becomes describable.

The next section positions VED within this framework.

12 The Position of VED

This volume introduced the structure of the *Differential Horizon* and reinterpreted divergence, renormalization, and the limits of modern physical theory.

Within this context, the position of VED (*Vortical Enclosure Dynamics*) can be clarified.

VED is not a unified theory.

It does not aim to integrate all interactions into a single framework.

VED does not describe the *un-generated* domain.

The *un-generated* is in principle indescribable. It is not an object of theory, but appears only as a boundary.

VED may be understood as a theory of the boundary of generation.

It describes the structure at the *Differential Horizon* and the behavior of theory in its vicinity.

VED does not close theory.

The *Differential Horizon* cannot be reached. Therefore, no theory can fully describe it.

This limit is not a deficiency,
but the condition under which theory can remain consistent.

VED does not replace existing theories.

Quantum field theory, relativity, and string theory remain valid within their respective domains.

VED shows the structure within which these theories are positioned.

Theory does not end here.
It simply cannot proceed further in the same form.

The *Differential Horizon* lies outside all theories. Yet all theories approach it.

Physics operates within the horizon.
Of what lies beyond, nothing can be said.

What lies beyond the boundary remains unsaid.

This volume has not provided a complete description of the world.

What has been reached is not an answer,
but a boundary.

13 Conclusion

This volume has traced the structure of divergence, renormalization, and the limits of modern physical theory.

It has introduced the *Differential Horizon* as a boundary between the *generated* and the *un-generated*.

It does not explain what lies beyond,
but clarifies where explanation ceases.

The *un-generated* domain cannot be described.
It is not an object of theory, but appears only as a boundary.

The boundary is not an object,
but a condition.

Divergence appears as theory approaches this boundary.
Renormalization converts that contact into a form that can be handled within theory.

These are not separate phenomena.

They are structural responses
to the same boundary.

Several questions remain open.

The precise mathematical structure of the *Differential Horizon*— its thickness, fluctuation spectrum, and relation to known geometric structures— awaits further development.

The connection between this framework and existing approaches to quantum gravity, holography, and information-theoretic methods remains to be made explicit.

The *Differential Horizon* is not a place that can be reached.

It marks the limit of what can be described.

Within the present framework, nothing further can be said.

Differential Horizon Map

References

- Yuhki Endoh. VED Vol. 1: Time and Space Generation, 2024a. Preprint, VED Research Group.
- Yuhki Endoh. VED Vol. 2: The Standard Model, 2024b. Preprint, VED Research Group.
- Yuhki Endoh. VED Vol. 3: Gravity and Cosmology, 2024c. Preprint, VED Research Group.
- Roger Penrose. Gravitational collapse and space-time singularities. *Physical Review Letters*, 14:57–59, 1965.
- Michael E. Peskin and Daniel V. Schroeder. *An Introduction to Quantum Field Theory*. Addison-Wesley, 1995.
- Max Planck. Über irreversible strahlungsvorgänge. *Sitzungsberichte der Königlich Preußischen Akademie der Wissenschaften*, pages 440–480, 1899.
- Joseph Polchinski. *String Theory, Vol. 1*. Cambridge University Press, 1998.
- Robert M. Wald. *General Relativity*. University of Chicago Press, 1984.
- Steven Weinberg. *The Quantum Theory of Fields, Vol. 1*. Cambridge University Press, 1995.
- Kenneth G. Wilson. The renormalization group and the epsilon expansion. *Physics Reports*, 12(2):75–199, 1974.

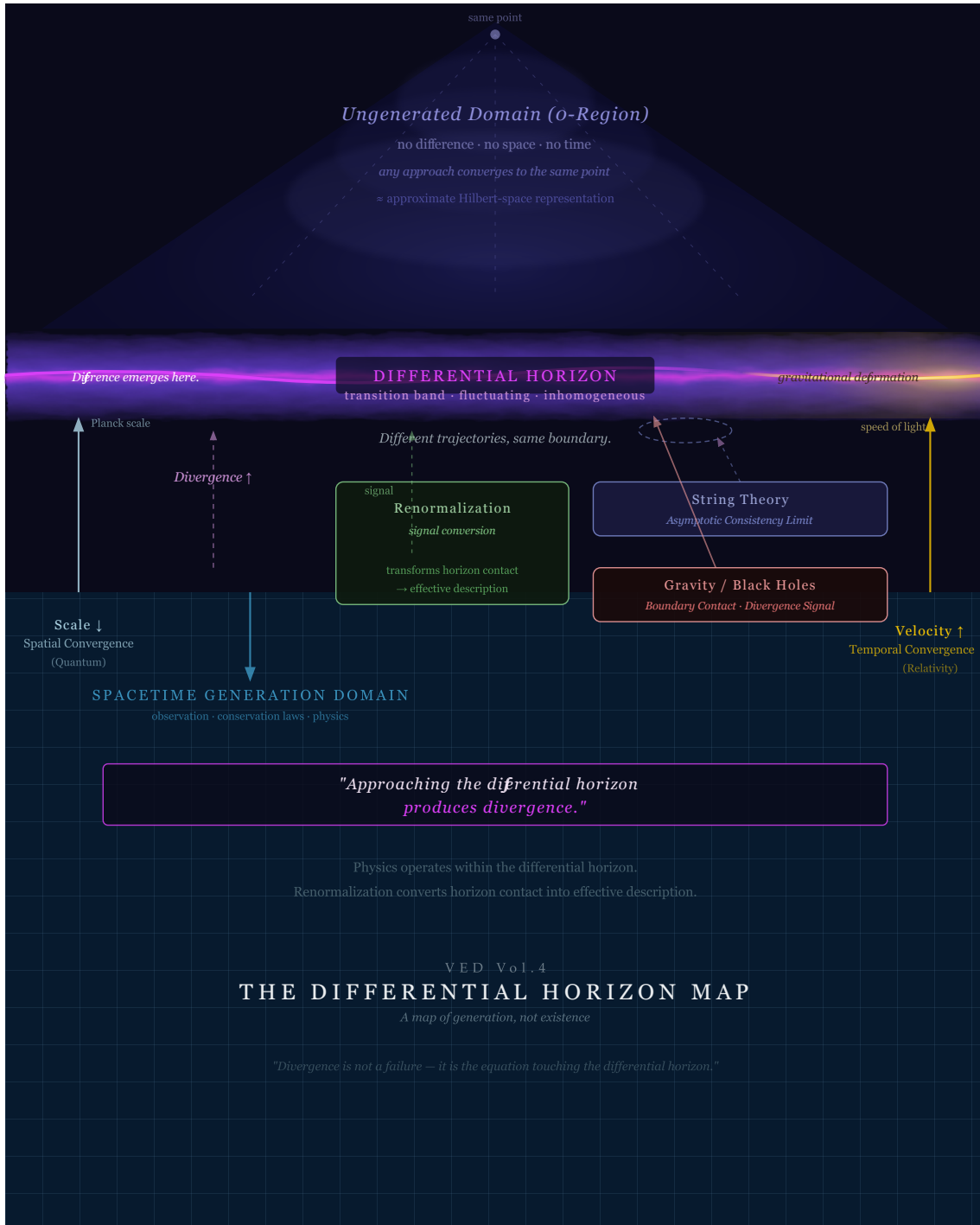


Figure 4: Differential Horizon map.